

Operating Systems

BCA — Semester IV — 2022 — Answer Sheet
Subject Code: CACS251

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. What is an operating system? Is "Multitasking" operating system better than "Batch" operating system? Justify your answer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is role of scheduler in operating system? Differentiate between short-term and long-term Scheduler.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by Race condition? Consider two concurrently running process P1 with statement S1 and P2 with statement S2. Suppose we require that S2 be executed only after S1 has completed. For this synchronization need, write a semaphore solution.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What are CPU scheduling criteria's? Consider below table of four processes under multi-level queue scheduling. Queue number denotes queue of the process. Priority of queue 1 is greater than queue 2. Queue 1 use Round Robin with time quantum $t=2\text{ms}$ and queue 2 use FCFS. Draw Gantt chart to illustrate their execution and calculate average waiting and turnaround time also.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is thrashing? Explain three memory management requirements?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Given five memory partition of 100KB, 500KB, 200KB, 300KB and 600KB (in order). How would the first fit, best fit and worst fit algorithm place processes of size 212KB, 417KB, 112KB, and 426KB (in order) in fixed-size partition? Which algorithm makes the most efficient use of memory?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. When programmed I/O is suitable than other I/O handling techniques? Explain different type System Threat.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Consider the following System: Process Allocation Max Available [A,B,C,D] [A,B,C,D] [A,B,C,D] P0 0 1 10 0 1 21 5 2 0 P1 0 0 01 7 5 0 P2 1 3 5 42 3 5 6 P3 0 6 3 20 6 5 2 P4 0 0 1 40 6 5 6 Answer the following questions using Banker's algorithm: a) What will be the contents of matrix need? b) Draw the safe sequence? c) If the request from P1 Arrives for (0, 4, 2, 0) can the request be granted immediately?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How second chance page replacement algorithm differ from FIFO page replacement policy? Explain the concept of belay's anomaly with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is clock synchronization? Explain how physical clock is synchronized by Cristian algorithm and logical clock is synchronized by Lamport's algorithm with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.